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NPS/002/009 – Specification for Cable Lugs and Connectors

1. Purpose

The purpose of this document is to detail the requirements of NEDL/YEDL in relation to mechanical cable lugs and connectors.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
DSS/005/014	1	Specification for mechanical Service Connectors
NE/MS-U-09		Technical Specification for Mechanical Connectors for use in Underground Cable joints
NE/MS-U-010		Technical Specification for Mechanical Connectors for use in LV Underground Cable joints

2. Scope

This document applies to a range of cable lugs, mechanical phase and neutral connectors to cover a wide range of cable combinations of both solid and stranded types which can be found on, or introduced to, the NEDL / YEDL distribution network.

Cable lugs are required to connect mains or service cables to various items of plant/switchgear.

3. Design Parameters

3.1 Conductor Types

Connectors and lugs shall be suitable for use on stranded copper, and solid aluminium, in both circular and sector-shaped cross section. The typical sizes of cables are as detailed in Appendix 1.

3.2 Design

The design must be such that the application of the connector/ lug results in rupture of metal oxide from the cable core to provide a true metal to metal contact.

3.3 Stress Relaxation Conductor Types

Connectors and lugs shall be so designed to allow for stress relaxation and differential thermal expansion under electrical loading.

3.4 Electrical Stability of the interface Connection

Aluminium connectors/lugs for use on copper conductors should be supplied with brass gauze for insertion between the aluminium connector/lug and the copper conductor to ensure low contact resistance during

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heat cycling. If other methods are proposed, details of testing and service experience to support the design should be provided.

3.5 Palm of Lug

The size of the palm shall reflect the current carrying capacity of the body of the connector.

3.6 Torque Feature

The design of the connector and lug shall include a torque related feature, which not only ensures that the correct clamping pressure is achieved and maintained, but can also be consistently reproduced between each screw in a multi screw connector/lug.

3.7 Conditions of Storage and Installation

During storage and after installation the lugs and connectors can be expected to be subjected to the full range of climatic conditions found within the UK.

3.8 Stud Size.

Stud size in the range of M6 to M16 is utilized on the majority of plant/equipment used within NEDL/YEDL

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4. References

4.1 External Documentation

Reference	Title
C 79	Type Approval Tests for Connectors and Terminations for aluminium Conductors of Insulated Power Cables
BS 4579	Specification for performance of mechanical and compression joints in electric cable and wire connectors. Compression joints in nickel, iron and plated copper conductors
BS 609	Code of practice for installation and inspection of un-insulated compression and mechanical connectors for power cables with copper or aluminium conductors

4.2 Internal Documentation

Reference	Title
Nil	

5. Definitions

Term	Definition
Nil	

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
R McMahon	Standards Engineer	R McMahon	03/04/06

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Bolden	Procurement Manager	G Bolden	23/05/06

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
D M Gazda	Underground Standards Manager	D M Gazda	23/05/06

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
Sean Johnson	DBD Administrator	Sean Johnson	31/03/06

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessments Manager	C Holdsworth	23/05/06

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	25/05/06

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Appendix 1 – Main Types of Cable found on the NEDL/YEDL System for which cable lugs may be required

11, 20 and 33kV Cables

- 95mm² - Solid Aluminium Core
- 185mm² - Solid Aluminium Core
- 300mm² - Solid Aluminium Core
- 0.1sq" Stranded Copper
- 0.2sq" Stranded Copper
- 0.3sq" Stranded Copper
- 0.4sq" Stranded Copper
- 0.5sq" Stranded Copper

3 – Core Waveform Cables with Sectoral Aluminium Cores

- 95mm² - Solid Aluminium Core & Copper Neutral Earth
- 185mm² - Solid Aluminium Core & Copper Neutral Earth
- 300mm² - Solid Aluminium Core & Copper Neutral Earth

Low Voltage Single Core Cable for Substations

- Armoured 480mm² Solid Sectoral Aluminium & Copper Neutral Earth
- Armoured 800mm² Stranded Copper Conductor & Copper Neutral Earth

Single and Three Phase CNE Service Cables

- 25mm² - Solid Aluminium Core & Copper Neutral Earth
- 35mm² - Solid Aluminium Core & Copper Neutral Earth

Single Phase SNE Service Cables

- 25mm² - Solid Aluminium Core & Copper Neutral & Earth
- 25mm² – Stranded Copper & Copper Neutral & Earth

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- 35mm² - Solid Aluminium Core & Copper Neutral & Earth

Earthing Conductors

- 6.0mm, (7/1.04mm), Green/Yellow PVC Insulated, 450/750V
- 16.0mm, (7/1.70mm), Green/Yellow PVC Insulated, 450/750V
- 25mm, (7/2.14mm)Green/Yellow PVC insulated 450/750V
- 35mm, (19/1.53mm) Green/Yellow PVC Insulated
- 70mm, (7/3.55mm) Green rated voltage 450/750V
- 120mm, (37/2.03mm) Green/Yellow PVC insulated

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Appendix 2 – Addendum to Supplier Requirements

The supplier shall provide with the tender full technical details of the equipment offered and shall indicate any divergence from these standards or specification.

To enable NEDL/YEDL to install the connectors/lugs in accordance with the manufacturer's recommendations the supplier should provide full instructions detailing the manufacturer's recommended method of installation of each product with due consideration, where applicable, to the range of application.

Where the products are considered to be suitable for a range of applications then the full scope of application within the range should be provided by the supplier for each tendered product.

No specialist tooling shall be required for the application of lug/connector.

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Appendix 3 – Logistical Requirements

To enable NEDL/YEDL to store the product(s) in accordance with the manufacturer's recommendations the supplier should provide details of the recommended storage environment with respect to each tendered product.

Details should be provided where relevant in respect to the minimum and maximum exposure levels, frequency of exposure and duration of exposure of the packaged item with respect to;

- | | | | |
|---|---------------------|---|-----------------------|
| * | Ambient temperature | * | Atmospheric corrosion |
| * | Humidity | * | Impact |
| * | Water | * | Vibration |
| * | Dust | * | Solar radiation |

The supplier shall ensure that each item is suitably packaged and protected to maintain the product and packaging as "fit for service" prior to installation, giving consideration to the potential for an outdoor storage environment. All packaging shall be sufficiently durable giving regard to the function, reasonable use and contents of the packaging. Where product packages tendered are made up of sub packages all the sub packages shall, unless varied by this specification, be supplied securely packaged together. Where items are provided in a bagged/boxed form, the material from which the bags are manufactured shall be capable of sustaining the package weight and resisting puncture by the materials within. The supplier shall submit at the time of tendering the details of the proposed packaging (i.e. materials composition and structure) to be used for each product. Where the supplier is unable to provide packaging suitable for outdoor storage then this should be stated at the time of tender.

Palletised goods shall be supplied on standard 1200mm x 1000mm pallets.

Clearly legible, easily identifiable, durable and unambiguous labelling shall be applied to each individual and where relevant multiple package of like products. Where product packages tendered are made up of sub packages each sub package shall be marked. As a minimum requirement the following shall be included;

- * Manufacturer's trademark or name
- * Supplier's trademark or name
- * Description of item
- * Date of packaging and/or batch number
- * NEDL/YEDL product code
- * Weight

The supplier shall submit at the time of tendering a sample of the proposed labelling for each product package type.